

**ATTACHMENT 6 – FACSIMILE TRANSMITTAL FROM MASSACHUSETTS
HIGHWAY DEPARTMENT TO GANNETT FLEMING, INC. DATED
SEPTEMBER 23, 1998 GIVING DIRECTION ON PROOF TEST LOAD FOR
ANCHORS
(3 pages)**



FACSIMILE TRANSMITTAL
MASSACHUSETTS HIGHWAY DEPARTMENT
ONE SOUTH STATION
BOSTON, MASSACHUSETTS 02110
FAX: 261-7552

DATE: 9-23-98
TIME: 8:35 PM

TO: RJ. Pomeroy

FAX NO. 380-7774

☐ STANDARD
☒ URGENT
☐ CONFIDENTIAL

FROM: EPB

FILE 29288	COMM	STRUC.	ACER	PECI
ES		CIVIL	ASEC	SAR
		GEOTH.	GMT	S&W

SUBJECT: ANCHOR BOLT TESTING

NO. OF PAGE(S) INCLUDING COVER SHEET 2

REQUESTED BY: _____ APPROVAL: _____

TRANSMITTED BY: _____ REQUESTED: _____

COMMENTS:
PLEASE INCORPORATE / APPROVE THESE
CHANGES BASED UPON LESSONS LEARNED IN TWT.

3.07 INSTALLATION OF ADHESIVE TYPE CEILING ANCHORAGE SYSTEM:

- A. Comply with adhesive anchor system manufacturer's installation instructions and recommendations.
- B. Use a non destructive procedure acceptable to the Engineer to locate concrete reinforcement, embedded utilities, conduits, etc. in tunnel structure.
- C. Use only drilling systems and drill sizes specifically recommended by the anchor manufacturer.
- D. Thoroughly clean bolt holes after drilling, with fresh, clean water or forced air as recommended by anchor system manufacturer.
- E. Do not cut concrete reinforcement or other embedded items. If reinforcing steel or other embedded items are encountered during drilling, abandon the hole and drill a new hole a maximum of 2 inches away from initial location shown. Patch abandoned holes with epoxy grout compatible with epoxy coating of reinforcing steel. Obtain approval of grout material from Engineer.

ALL

- F. After chemical adhesive has cured, proof test anchors to 125 percent of their design load per ASTM E 448, unconfined static tests for tension and shear, ~~using procedures outlined in Subparagraph 3.07F1 below.~~ Record results of proof tests and submit to Engineer. Clearly mark and abandon anchors which fail test. Proof test procedure must be submitted to and reviewed by Engineer prior to installation. ~~REPAIRS TO.....~~

1. Anchor Bolt Testing Procedure

- DELETE**
- a. Step I: Test at least 20 percent of each lot of 100 anchor bolts installed. ~~Actual anchors to be tested will be randomly selected by the Engineer. If no anchor fails, continue testing 20 percent of each lot of 100 anchors. Should any anchor fail during the 20 percent testing, proceed to Step II.~~
 - b. Step II: Test an additional 50 anchors from the lot in which failure occurred, including (if available) 15 anchors of the anchor size that failed. Should none of these 50 anchors fail, proceed to Step III. If only one anchor fails, test all remaining anchors in this lot. If none of the remaining anchors fail, proceed to Step III for the next lot. If any of the remaining anchors fail, proceed to Step IV for the next lot.
 - c. Step III: Test 20 percent of the next 100 anchors. If no anchor fails, go to Step I for the next lot. If any anchor fails, go to Step II.
 - d. Step IV: Test 50 of the next 100 anchors. If no anchor fails, proceed to Step III. If any anchor fails, test the remaining 50 anchors in this lot. If none of these 50 anchors fail, return to Step III for the next lot. If any of the remaining 50 anchors fail, repeat Step IV for the next lot.

Repairs to concrete surfaces required due to Anchorage System testing shall be made, at no additional cost to the Department. Repair materials must be acceptable to the Engineer.

- G. Under no circumstances shall an anchor be installed within the embedment length of the face of the electric pull boxes or similar embedments.
- H. Engineer may conduct spot checks on anchors.
- I. Fully torque bolt to manufacturers specifications.

4.02 BASIS OF PAYMENT